PATTERN_SIGNED_RR: signed lift mass per 140Q fibre vs MF2

2026-10-24 (America/Chicago)

RESULTS: PATTERN_SIGNED_RR (Map-PSR1)

PREREG sha256 98a1db170d68f74d1a7991050e417959430e498172888fb378ab3ac05dfd74d8. Hashed after explore (R=150,300) and before test (fresh R=400; logs/test.log).

fibre	ρ_res (MF2)	ρ_mod (signed lift mass, R=400)
0	1/32	~$4\cdot10^{-11}$
1	1/4	0
2	3/32	~$8\cdot10^{-11}$
3	0	1/2
4	7/32	~$4\cdot10^{-11}$
5	1/4	0
6	5/32	0
7	0	1/2

- Supports are **disjoint**: {3,7} vs {0,1,2,4,5,6}. TV ≈ 1.
- PQ1 is not escaped: fibres 1, 5 and 6 are exactly 0.
- Non-vacuous.

Verdict: FAIL (Category B numerically; the support statement on 1, 5, 6 is Category A by PS1a). Bridge: **NO.** ## NOTES: PATTERN_SIGNED_RR ## Reconstruction (written) - w(v) = ½(sgn 5n − sgn(−n−21k)), from rev8 l.156–158 (c1 = (−1,1), c2 = (−4,1)). - Points ν = α+v. Phase e(B(ν,β)), β ∈ {0,−a}, B(x,y) = x^T[[2,7],[7,7]]y (Q = ½B). - I use the signed cone-weighted lattice content, i.e. the H_s-type indefinite theta per label. I do not use the full Siegel-lift integral kernel of §sec:lift (geodesic integral), which is not reconstructed here. This is a written but partial reconstruction of u_{α,β}.

Lemma PS1

- **PS1a (Category A).** Any weighting of lattice points that is binned by 140·Q(v) mod 8 (signed, complex, cone-weighted, or the full kernel too) is supported inside the unsigned support of PQ1, {0,2,3,4,7}. So fibres 1, 5 and 6 are always empty. **No**

refinement of the coefficient weighting can reach MF2 while the fibre is 140Q mod 8.

- **PS1b (Category B).** With the signed cone weight, the L′-coset labels (fibres {0,2,4}) cancel to relative size ~10^{-11}. Only the a-coset survives, so the signed mass sits on {3,7}, exactly the complement of MF2's support.
  - The likely mechanism is w(−v) = −w(v) together with the label symmetry α ↔ −α on L′. This is not proved here.

Observation (post hoc, not a claim)

The Resonant signed mass lives exactly on MF2's empty fibres {3,7}. This sharpens the PQ1 observation. A complement map is still not registered, because there is no written reason independent of ρ_res.

Next pattern step

The obstruction is now in the **fibre coordinate** (140Q mod 8), not in the weights. The pattern-forced alternative is a different written fibre coordinate on the Resonant side. Two candidates: - B(ν,a) mod 8-type: rev8 l.896 gives B(ν,a) = Re θ/10. - The T-phase class Q(α)+B(α,w_c) from the written T-law (l.272–275), reduced to Z/8. Either should be preregistered only if a written reason ties it to an 8-fibre structure. Otherwise ask Arvin for the intended Resonant fibre coordinate.